

Combining the Power of J Coupling and DP4 Analysis on Stereochemical Assignments: The J -DP4 MethodsNicolás Grimblat,[†] José A. Gavín,[‡] Antonio Hernández Daranas,^{*,‡,§} and Ariel M. Sarotti^{*,†}[†]Instituto de Química Rosario (IQUIR, CONICET-UNR) and Facultad de Ciencias Bioquímicas y Farmacéuticas, Universidad Nacional de Rosario, Suipacha 531, S2002LRK Rosario, República Argentina[‡]Instituto Universitario de Bio-Organica Antonio González, Universidad de La Laguna, 38206 Tenerife, Spain[§]Instituto de Productos Naturales y Agrobiología del CSIC (IPNA-CSIC), La Laguna, 38206 Tenerife, Spain

Supporting Information

ABSTRACT: A systematic study to include $^3J_{\text{HH}}$ couplings into DP4 formalism (J -DP4) led to the development of three alternative strategies. The dJ -DP4 (direct) approach involves a new DP4-like equation including an additional probability term given by $^3J_{\text{HH}}$. The iJ -DP4 (indirect) approach explores the original DP4 method with a restricted conformational search. Despite both strategies performing better than DP4, their combined use (iJ/dJ -DP4) provided the best results, with a 2.5-fold performance improvement at similar or lower computational cost.



A complete knowledge of the stereochemical features of organic molecules is essential to understand their chemical and biological properties. This goal is usually achieved from analysis of NMR data. Despite continuous advances in spectroscopic techniques,¹ this task is still challenging, as is evident from the large number of erroneously assigned structures.²

Traditionally, relative configurations have been attained by the use of angular information deduced from H–H scalar coupling constants ($^3J_{\text{HH}}$) and/or distance restraints obtained from the nuclear Overhauser effect (NOE). Further development has been achieved by incorporation of long-range heteronuclear coupling constants ($^2,3J_{\text{CH}}$) in the so-called J -based configuration analysis, which has been widely used to determine the relative configurations in acyclic or potentially flexible cyclic molecules.³ More recently, residual dipolar couplings (RDCs) and residual chemical shift anisotropy (RCSA) have been validated as powerful elucidation tools.⁴

On the other hand, theoretical calculation of spectroscopic properties such as NMR chemical shifts and scalar couplings by means of computational chemistry techniques is increasingly easy and reliable.⁵ This has made the comparison between experimental and computed values for a set of putative candidates a powerful alternative to assist in the structural assignment of organic molecules.^{5,6} Among these approaches, the DP4 probability stands out. This method is formulated on the basis that Bayes's theorem can be used to estimate the probability that the chosen solution is correct.⁷

In our opinion, one of the main sources of error in DP4 analysis is related to the capabilities of the level of theory employed (B3LYP/6-31G**//MMFF) to adequately represent the real conformational distribution of the target molecule.⁸ Thus, flexible compounds usually generate larger errors and therefore inferior results in predictions. Improved methods such as DP4+ relies on the use of more refined geometries achieved

by the inclusion of an additional geometrical optimization step as well as a better treatment of the GIAO NMR calculations. More recently, Goodman and co-workers reported an updated DP4 method (DP4.2) by computing the NMR shifts at the higher mPW1PW91/6-311G* level and the relative energies for Boltzmann analysis at the M06-2X/6-31G** level.^{9,10} Despite both methods outperforming the original DP4, the improvement was accompanied by a higher computational cost.^{9,10} Hence, we foresee the development of a new computational toolbox able to offer better results than DP4 while keeping the associated computational cost to a minimum.

Vicinal couplings ($^3J_{\text{HH}}$) are readily available from a standard ^1H NMR experiment, and the information they provide is extremely valuable for conformational and stereochemical studies. Thus, extensive effort has been made in the field of quantum-based calculations of J .¹¹ Those results have been employed in configurational determination of natural products.¹² However, it is striking that such a crucial stereochemical indicator has never been incorporated into DP4. In consequence, we decided to explore this possibility.

According to our understanding, the information provided by $^3J_{\text{HH}}$ can be integrated into the standard DP4 formalism in two different ways, namely, direct (dJ -DP4) or indirect (iJ -DP4). In dJ -DP4, δ_{H} , δ_{C} , and $^3J_{\text{HH}}$ are computed at the DFT level to be later correlated with experimental values using a modified DP4 equation, which must account for the probability of a certain difference between calculated and experimental $^3J_{\text{HH}}$ values. Alternatively, in the iJ -DP4 approach, the experimental $^3J_{\text{HH}}$ values are used to introduce geometric restrictions into the conformational sampling. Thus, just compatible conformations are kept for NMR chemical shift calculations, which are then

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further evaluated using standard DP4 analysis. In both ways, selection of the candidate structure reduces dependence on the theoretically computed energy values used to calculate the relative populations of the possible conformations. Figure 1 illustrates the two approaches under study.

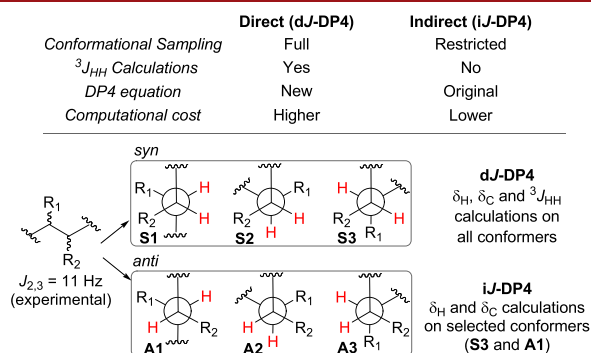


Figure 1. Schematic representation of iJ- and dJ-DP4.

To explore the feasibility of both approaches, we thoroughly investigated a selected set of 69 examples, featuring a wide variety of molecular complexity and conformational freedom (Figure 2). To compare them with the original DP4 procedure, the conformational searches were done at gas phase using the MMFF force field as implemented in Spartan or Macromodel. All conformers found within 5 kcal/mol from the global minima were next submitted to NMR calculations (δ_{H} , δ_{C} , and $^3J_{\text{HH}}$) at the B3LYP/6-31G** level using GIAO. $^3J_{\text{HH}}$ values were calculated considering just Fermi contact term (FC) following previous reports.^{11a}

The dJ-DP4 probability of isomer *i* (out of *m* isomers) being the correct structure involves modification of the original DP4

formalism (first half of eq 1) by including the new probability term given by $^3J_{\text{HH}}$ (second half of eq 1).

$$dJ\text{-DP4}(i) = \frac{\prod_{k=1}^N \left(\frac{1 - T^{\nu}(\delta_{\text{H},k}^i - \delta_{\text{exp},k})/\sigma_{\delta,\text{H}}}{1 - T^{\nu}(\delta_{\text{H},k}^j - \delta_{\text{exp},k})/\sigma_{\delta,\text{H}}} \right) \prod_{k=1}^N \left(\frac{1 - T^{\nu}(J_{\text{H},k}^i - J_{\text{exp},k})/\sigma_{J,\text{H}}}{1 - T^{\nu}(J_{\text{H},k}^j - J_{\text{exp},k})/\sigma_{J,\text{H}}} \right)}{\sum_{j=1}^m \left[\prod_{k=1}^N \left(\frac{1 - T^{\nu}(\delta_{\text{H},k}^j - \delta_{\text{exp},k})/\sigma_{\delta,\text{H}}}{1 - T^{\nu}(\delta_{\text{H},k}^i - \delta_{\text{exp},k})/\sigma_{\delta,\text{H}}} \right) \prod_{k=1}^N \left(\frac{1 - T^{\nu}(J_{\text{H},k}^j - J_{\text{exp},k})/\sigma_{J,\text{H}}}{1 - T^{\nu}(J_{\text{H},k}^i - J_{\text{exp},k})/\sigma_{J,\text{H}}} \right) \right]} \quad (1)$$

The dJ-DP4 probability can be defined knowing the degrees of freedom (ν) and standard deviation (σ) of the chemical shift and $^3J_{\text{HH}}$ terms at a given level of theory. In an effort to understand the effect of including $^3J_{\text{HH}}$ in DP4 equations, the $[\nu, \sigma]$ terms corresponding to δ_{H} and δ_{C} were kept as in the original reference.⁷ On the other hand, the $^3J_{\text{HH}}$ $[\nu, \sigma]$ values were estimated as 0.992 and 3.06, respectively, by correlating the experimental and calculated *J* data using our test data set. To build up *t* distributed series, the calculated $^3J_{\text{HH}}$ values were first scaled according to $J_s = (J_{\text{calc}} - b)/m$. We chose a generalized linear scaling procedure (using fixed *m* and *b* values resulting from a large data set) as followed by Gonella and co-workers during the development of DiCE.¹³

From the resultant plot of J_{calc} versus J_{exp} (Figure 3) we estimated the scaling parameters as $b = -0.14$ and $m = 0.95$. Despite the wide data dispersion ($R^2 = 0.85$), the error series followed a *t* distribution, validating the use of eq 1. In some cases, there were significant differences between experimental and calculated values, which was the result of including unsuitable conformations with relative stability being overestimated by DFT energies (for instance, featuring intramolecular H-bonding). Removing these spurious data points (41 errors higher than 3 Hz) afforded a considerably higher R^2 value (0.92), though the scaling parameters hardly changed ($b = -0.14$, $m = 0.95$). In addition, our slope was in agreement with the value found by Bally and Rablen at the B3LYP/6-31G(d,p)+1s level (0.912).^{11a}

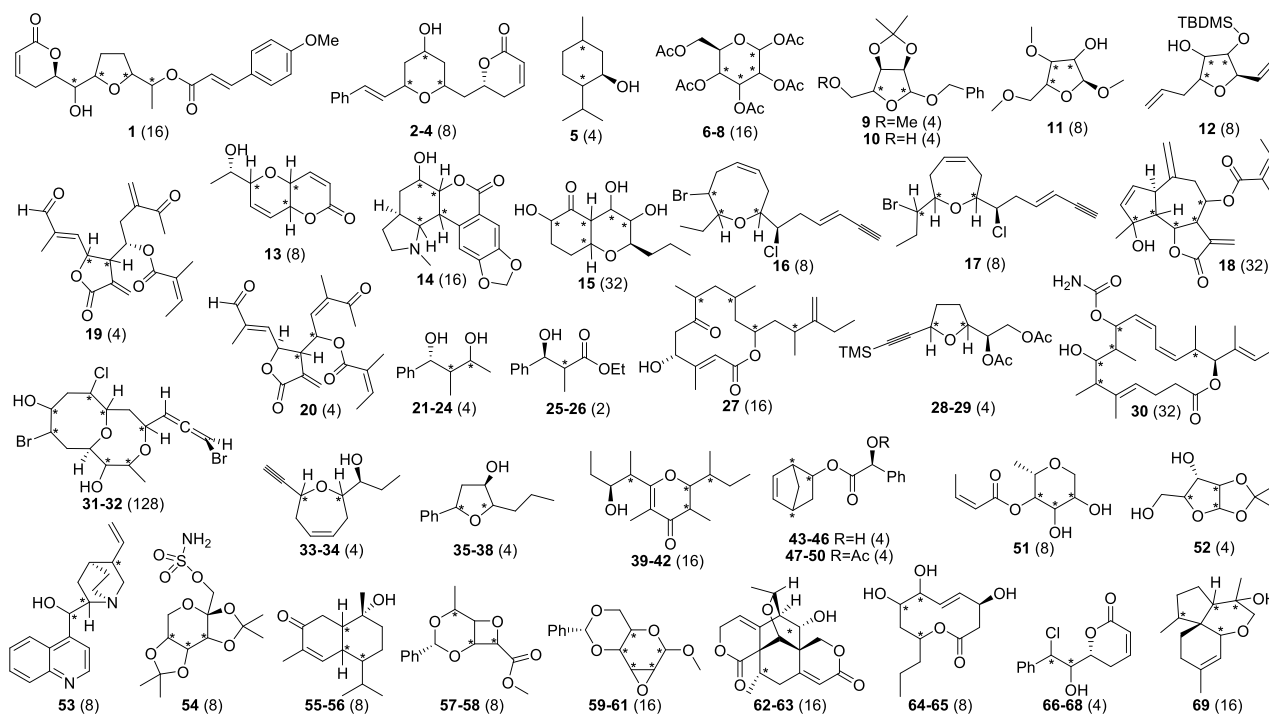


Figure 2. Test set of 69 compounds. The number of possible candidates generated by varying the configurations at the carbons marked with an asterisk is provided in parentheses.

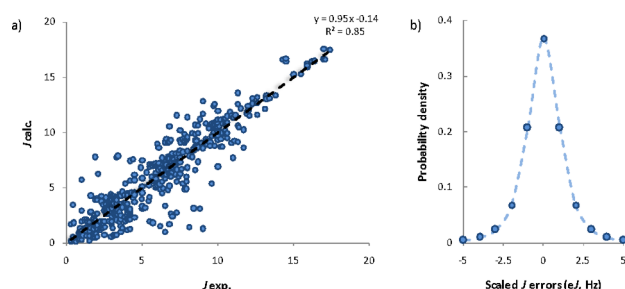


Figure 3. (a) Correlation plot of J_{calc} vs J_{exp} . (b) Probability density of scaled J errors ($eJ = J_{\text{scaled}} - J_{\text{exp}}$) computed at the B3LYP/6-31G**//MMFF level of theory.

With our test data set, the standard DP4 method afforded a 75% correct classification (Figure 4), whereas the dJ-DP4 probability gave a 96% correct classification, which was a significant enhancement. Moreover, just 39% of the cases were successfully predicted by DP4 with high confidence (>95%), clearly lower than the 70% achieved with dJ-DP4. To facilitate quantification of the improvement allowed by inclusion of $^3J_{\text{HH}}$, we envisaged an arbitrary scoring system depending on the confidence achieved at a certain assignment (2 points for a correct assignment made with high confidence (>95%), 1 for a correct assignment with a more modest confidence (>85%), 0 to a correct but low-confidence assignment, and -1 to a wrong

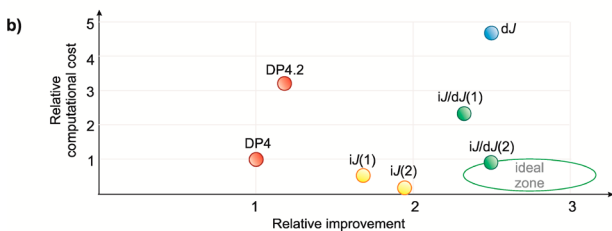
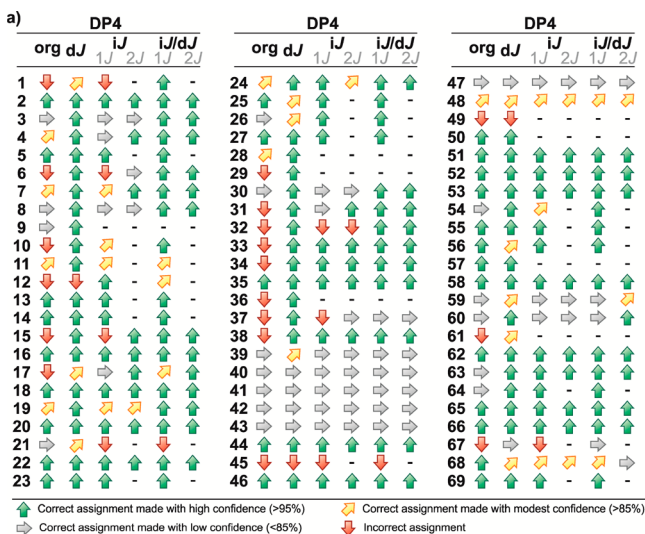


Figure 4. (a) Results obtained with the original DP4 (org) and J-DP4 (dJ, iJ, and iJ/dJ) formalisms with our test set of 69 examples. The iJ-DP4 and iJ/dJ-DP4 methods were evaluated using one (1J) or two (2J) experimental $^3J_{\text{HH}}$ values to constrain the conformational sampling (indicated in the number in parentheses in panel b). (b) Overall performance improvement and computational cost (relative to original DP4 method). The relative improvement of DP4.2 was estimated using our scoring system with the data presented in the original reference.¹⁰

assignment). Within these boundaries, the relative score of dJ-DP4 (78) was 2.5 times higher than that achieved with DP4 (31). In several examples, a wrong assignment made by DP4 was reverted by dJ-DP4 (e.g., compounds 1, 6, 10, 15, 29, 31–34, 37, and 61). On the other hand, in just 3 cases (compounds 25, 56, and 68) dJ-DP4 afforded more modest results than DP4, though the right isomers were still identified as the most probable ones. To better illustrate the usefulness of the dJ-DP4 approach, Figure 5a shows the results obtained for the four

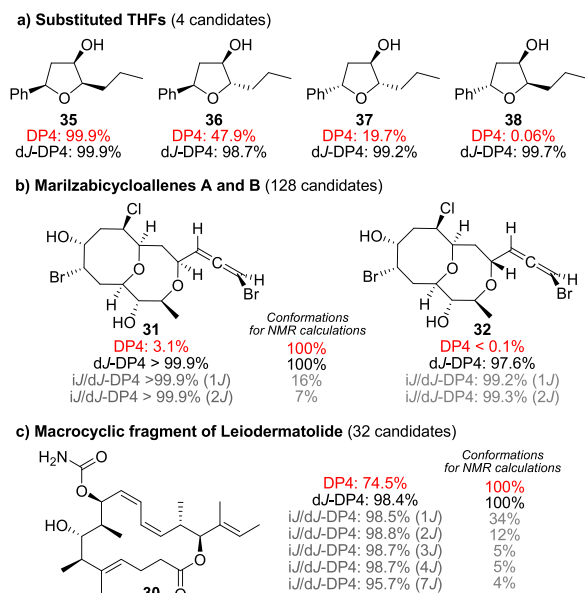


Figure 5. Three representative case studies.

THFs 35–38, whose assignment by computational methods has proven to be highly challenging via DP4 given the flexibility of five-membered rings.¹⁴ Whereas DP4 correctly assigned just isomer 35, using our dJ-DP4 approach the four isomers could be successfully elucidated in high probability.

The indirect strategy (iJ-DP4) was also evaluated. This approach considers the subset of candidates compatible with experimental $^3J_{\text{HH}}$ values. Because the experimental $^3J_{\text{HH}}$ value is the result of averaging the contributions made by the different conformers, large (i.e., >8–9 Hz) or low (i.e., <4 Hz) $^3J_{\text{HH}}$ values were used to constrain the conformational sampling (the protons are either almost coplanar or perpendicular, respectively). On the other hand, intermediate values ($4 < ^3J_{\text{HH}} < 8$ Hz) could arise from the averaging of large and small coupling values, preventing their use to guide the conformational search. This is one of the main practical limitations of iJ-DP4 and is the reason why some systems under study could not be evaluated by this method (compounds 9, 28, 29, 36, 49, 50, 57, and 61). In the remaining 61 cases, we found that using one experimental $^3J_{\text{HH}}$ value is enough to guide the conformational search and afford good results in terms of DP4 performance (1.7-fold improvement at half the computational cost, Figure 4). The use of a second $^3J_{\text{HH}}$ value has an even more pronounced effect, both in scoring (2 times higher) and computational cost (~80% lower). In summary, iJ-DP4 affords generally better results than DP4 at a significantly reduced computational cost. This could be extremely helpful to study large molecules that require long calculation times.

According to our results, the dJ-DP4 method provided better performance than iJ-DP4, which in turn had the advantage of

being much more affordable in terms of computational cost. Hence, we anticipated that mixing both approaches would afford better results. In this new *ij*/*dj*-DP4 alternative, relevant $^3J_{\text{HH}}$ couplings are used to guide conformational sampling and the resulting conformations are employed to calculate their NMR parameters for *dj*-DP4 analysis. To avoid data redundancy, $^3J_{\text{HH}}$ values used to constrain the conformational search were neglected in the *dj*-DP4 calculations. To our delight, we found that this merged method afforded excellent results, keeping the high scoring achieved by *dj*-DP4 at a fraction of the computational cost as in *ij*-DP4. Thus, using one $^3J_{\text{HH}}$ value to restrict conformational sampling, *ij*/*dj*-DP4 gave 2.3 times higher scoring than the original DP4 approach at about twice the computational cost. Using two $^3J_{\text{HH}}$ values afforded even higher performance (2.5 times) with a lower computational cost than DP4 (Figure 4b).

The power of our *ij*/*dj*-DP4 approach can be illustrated in the stereoassignment of marilzabicycloallenes A and B, two complex natural products featuring a flexible [5.5.1]bicyclotridecane ring system (Figure 5b).¹⁵ Using the standard DP4 treatment, the right structures (31 and 32, respectively) were identified in very low probabilities from the set of 128 different possible candidate stereoisomers (3.1% and 0.1%) and in high confidence (>95%) a wrong isomer. Including the experimental coupling values via *dj*-DP4 drastically improved the results (>99.9% and 97.6%), but with a concomitant increase in the computational cost. Moreover, by restricting the conformational samplings using one or two vicinal couplings, the number of suitable conformations used for DFT calculations was significantly reduced (84% and 93%, respectively). The resulting *ij*/*dj*-DP4 probabilities still afforded the right assignment by *dj*-DP4 but with a fraction of the overall computational cost.

The *ij*/*dj*-DP4 approach can be employed with as many suitable *J* values as possible, but generally, the inclusion of more than 3–4 nontrivial $^3J_{\text{HH}}$ does not produce further improvements. For instance, in the case of the macrocyclic fragment of leiodermatolide (30, Figure 5c),¹⁶ restricting the conformational sampling with 2 and 3 experimental $^3J_{\text{HH}}$ values resulted in a ~8 and ~20-fold reduction, respectively, in the number of conformations employed for further NMR calculations while confirming the right assignment made by *dj*-DP4 (98.8% and 98.7%, respectively). Including up to 7 additional couplings (to strengthen the constraint) resulted in a minor effect on the resulting number conformations, with no significant effect on the *ij*/*dj*-DP4 values.

In conclusion, we have carried out a systematic study to incorporate the valuable information provided by $^3J_{\text{HH}}$ coupling into the DP4 formalism. The three alternatives herein developed (*dj*-DP4, *ij*-DP4, and *ij*/*dj*-DP4) resulted in an important improvement in DP4 performance. Among them, the *ij*/*dj*-DP4 strategy was noteworthy, affording a significant improvement in DP4 performance (up to 2.4 times higher) at the same (or even lower) computational cost. Therefore, this method offers a convenient strategy for a fast and accurate stereoassignment of complex organic compounds. As a final remark, it should be emphasized that the reliability of *J*-DP4 calculations still depends on the accuracy provided by the B3LYP/6-31G**//MMFF level to effectively account for the real conformational distribution. The inclusion of *J* values to constrain the conformational sampling allowed solving, at least in part, this drawback. Nevertheless, systems featuring high flexibility and/or possible intramolecular H-bonding interactions remain a great challenge, and special care should be taken in those cases when

analyzing the *J*-DP4 results. Finally, in an effort to facilitate the *J*-DP4 calculations, an Excel file is provided free of charge at sarotti-nmr.weebly.com, iquir-conicet.gov.ar/Sarotti-NMR, and the digital CSIC repository (dx.doi.org/10.20350/digitalCSIC/8634).

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.orglett.9b01193.

Instructions for *J*-DP4 calculations; full list of compounds used in the test set (with references); computational details; experimental and calculated NMR data of all compounds under study; and values for DP4 and *J*-DP4 methods (PDF)

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Notes

The authors declare no competing financial interest.

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■ REFERENCES

- (1) Williams, A.; Martin, G.; Rovnyak, D. *Modern NMR Approaches to the Structure Elucidation of Natural Products: Vol. 2: Data Acquisition and Applications to Compound Classes*; Royal Society of Chemistry, 2016.
- (2) (a) Chhetri, B. K.; Lavoie, S.; Sweeney-Jones, A. M.; Kubanek, J. *Nat. Prod. Rep.* **2018**, 35, 514–531. (b) Nicolaou, K.; Snyder, S. A. *Angew. Chem., Int. Ed.* **2005**, 44, 1012–1044.
- (3) (a) Matsumori, N.; Kaneno, D.; Murata, M.; Nakamura, H.; Tachibana, K. *J. Org. Chem.* **1999**, 64, 866–876. (b) Napolitano, J. G.; Gavín, J. A.; García, C.; Norte, M.; Fernández, J. J.; Hernández Daranas, A. *Chem. - Eur. J.* **2011**, 17, 6338–6347.
- (4) (a) Nath, N.; Schmidt, M.; Gil, R. R.; Williamson, R. T.; Martín, G. E.; Navarro-Vázquez, A.; Griesinger, C.; Liu, Y. *J. Am. Chem. Soc.* **2016**, 138, 9548–9556. (b) Liu, Y.; Sauri, J.; Mevers, E.; Pecuh, M. W.; Hiemstra, H.; Clardy, J.; Martín, G. E.; Williamson, R. T. *Science* **2017**, 356, No. eaam5349. (c) Troche-Pesqueira, E.; Anklin, C.; Gil, R. R.; Navarro-Vázquez, A. *Angew. Chem., Int. Ed.* **2017**, 56, 3660–3664. (d) Liu, Y.; Navarro-Vázquez, A.; Gil, R. R.; Griesinger, C.; Martín, G. E.; Williamson, R. T. *Nat. Protoc.* **2019**, 14, 217–247.
- (5) For leading reviews, see: (a) Bifulco, G.; Dambrosio, P.; Gomez-Paloma, L.; Riccio, R. *Chem. Rev.* **2007**, 107, 3744–3779. (b) Lodewyk, M. W.; Siebert, M. R.; Tantillo, D. J. *Chem. Rev.* **2012**, 112, 1839–1862. (c) Grimblat, N.; Sarotti, A. M. *Chem. - Eur. J.* **2016**, 22, 12246–12261.
- (6) (a) Smith, S. G.; Goodman, J. M. *J. Org. Chem.* **2009**, 74, 4597–4607. (b) Zanardi, M. M.; Sarotti, A. M. *J. Org. Chem.* **2015**, 80, 9371–

9378. (c) Navarro-Vazquez, A.; Gil, R. R.; Blinov, K. *J. Nat. Prod.* **2018**, *81*, 203–210. (d) Kutateladze, A. G.; Reddy, D. S. *J. Org. Chem.* **2017**, *82*, 3368–3381.
- (7) Smith, S. G.; Goodman, J. M. *J. Am. Chem. Soc.* **2010**, *132*, 12946–12959.
- (8) Domínguez, H. J.; Crespín, G. D.; Santiago-Benítez, A. J.; Gavín, J. A.; Norte, M.; Fernández, J. J.; Daranas, A. H. *Mar. Drugs* **2014**, *12*, 176–192.
- (9) Grimblat, N.; Zanardi, M. M.; Sarotti, A. M. *J. Org. Chem.* **2015**, *80*, 12526–12534.
- (10) Ermanis, K.; Parkes, K. E. B.; Agback, T.; Goodman, J. M. *Org. Biomol. Chem.* **2017**, *15*, 8998–9007.
- (11) (a) Bally, T.; Rablen, P. R. *J. Org. Chem.* **2011**, *76*, 4818–4830. (b) Kutateladze, A. G.; Mukhina, O. A. *J. Org. Chem.* **2014**, *79*, 8397–8406.
- (12) (a) Sarotti, A. M. *Org. Biomol. Chem.* **2018**, *16*, 944–950. (b) Cen-Pacheco, F.; Rodríguez, J.; Norte, M.; Fernández, J. J.; Hernández Daranas, A. *Chem. - Eur. J.* **2013**, *19*, 8525–8532. (c) Reddy, D. S.; Kutateladze, A. G. *Org. Lett.* **2016**, *18*, 4860–4863.
- (13) Xin, D.; Jones, P.-J.; Gonnella, N. C. *J. Org. Chem.* **2018**, *83*, 5035–5043.
- (14) Kang, B.; Mowat, J.; Pinter, T.; Britton, R. *Org. Lett.* **2009**, *11*, 1717–1720.
- (15) Gutiérrez-Cepeda, A.; Fernández, J. J.; Norte, M.; Souto, M. L. *Org. Lett.* **2011**, *13*, 2690–2693.
- (16) Paterson, I.; Dalby, S. M.; Roberts, J. C.; Naylor, G. J.; Guzmán, E. A.; Isbrucker, R.; Pitts, T. P.; Linley, P.; Divlianska, D.; Reed, J. K.; Wright, A. E. *Angew. Chem., Int. Ed.* **2011**, *50*, 3219–3223.